ACADEMIC CENTRAL

PRACTICE EXAMINATION

GRADE 11

MECHANICAL TECHNOLOGY

THEORY PAPER

MARKS: 100 **TIME: 2 hours**

Topics: safety and tools, materials and heat treatment, forces (stress and strain, beams, resultants), systems and control (gears, belts, hydraulics), maintenance and joining methods. Suitable for Automotive, Fitting and Machining, and Welding and Metalwork learners.

Written by Academic Central for practice, following the CAPS curriculum for Grade 11. It is not a Department of Basic Education examination paper.

INSTRUCTIONS AND INFORMATION

1. Answer ALL the questions. Show ALL formulae, substitutions and units in calculations. Round answers to TWO decimal places.
2. A non-programmable calculator may be used.

QUESTION 1: MULTIPLE CHOICE

1.1 The heat-treatment process that reduces the brittleness of hardened steel is …

A annealing. B tempering.
C normalising. D case hardening.

1.2 The instrument used to measure wear in an engine cylinder bore is a …

A dial bore gauge. B torque wrench.
C feeler gauge. D compression tester.

1.3 A driver gear with 20 teeth turns at 900 r/min and drives a gear with 60 teeth. The driven gear turns at …

A 300 r/min. B 2 700 r/min.
C 450 r/min. D 1 800 r/min.

1.4 A welding defect where weld metal flows onto the parent metal without fusing with it is called …

A overlap. B porosity.
C undercut. D spatter.

1.5 The carbon content of mild steel is about …

A 0,1–0,3%. B 0,8%.
C 1,4%. D 3–4%.

(5 × 2)
[10]

QUESTION 2: SAFETY AND TOOLS

2.1 Give FOUR safety rules when working on a centre lathe. (4)

2.2 State the use of: (a) a torque wrench; (b) a cylinder leakage tester; (c) a compression tester. (3)

2.3 On a metric micrometer, the 12 mm line and one half-millimetre line are visible on the sleeve, and line 27 on the thimble is in line with the datum line. Give the reading. (2)

2.4 Give FOUR safety precautions when using oxy-acetylene gas welding equipment. (4)

2.5 Give TWO reasons why acetylene cylinders must be stored and used upright. (2)

[15]

QUESTION 3: MATERIALS

3.1 For each heat-treatment process, describe how it is done and state its purpose: (a) hardening; (b) tempering; (c) annealing; (d) normalising. (8)

3.2 What is case hardening? Give ONE component that is case hardened. (3)

3.3 Explain how (a) a file test and (b) a sound test help to identify metals in the workshop. (4)

[15]

QUESTION 4: FORCES

4.1 A round steel bar 1,5 m long with a diameter of 30 mm carries a compressive load of 80 kN. Young's modulus is 210 GPa. Calculate: (a) the stress in MPa; (b) the strain; (c) the change in length in mm. (7)

4.2 The beam AB in FIGURE 4.2 is 8 m long and simply supported at A and B. It carries point loads of 30 kN and 50 kN, and a uniformly distributed load of 5 kN/m over its whole length. Calculate the reactions at A and B. (7)

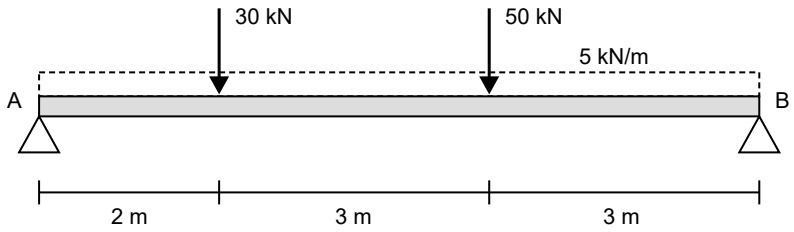


FIGURE 4.2

4.3 Three forces act at a point. Angles are measured anticlockwise from the positive x-axis: 120 N at 30°, 80 N at 135° and 50 N at 270°. Calculate the magnitude and direction of the resultant. (8)

4.4 Explain what is meant by the *elastic limit* of a material. (3)

[25]

QUESTION 5: SYSTEMS AND CONTROL

5.1 Gear A (24 teeth) turns at 1 200 r/min and drives idler gear B (36 teeth), which drives gear C (72 teeth). Calculate: (a) the speed of gear C; (b) the velocity ratio of the gear train. (c) State the function of the idler gear. (6)

5.2 A driver pulley with a diameter of 150 mm turns at 1 440 r/min and drives a pulley with a diameter of 400 mm. Calculate: (a) the speed of the driven pulley; (b) the belt speed in m/s. (5)

5.3 In a hydraulic press, a force of 300 N acts on a small piston with a diameter of 25 mm. The large piston has a diameter of 200 mm. Calculate: (a) the fluid pressure in kPa; (b) the force exerted by the large piston. (6)

5.4 Give THREE advantages of a V-belt drive over a flat-belt drive. (3)

[20]

QUESTION 6: MAINTENANCE AND JOINING METHODS

6.1 Distinguish between preventive maintenance and corrective maintenance. (4)

6.2 Name THREE welding defects (not overlap). For each, give ONE cause and ONE way to prevent it. (6)

6.3 Name FIVE methods used to stop nuts and bolts from working loose. (5)

[15]

TOTAL: 100

END OF QUESTION PAPER